



PMSETU ToT — INFORMATION TECHNOLOGY | DAY 8

Network & Wi-Fi Security Essentials

Secure Administrative Access • LAN & Layer 2 Security • Endpoint Security • Wi-Fi Security

Friday, 11 September 2026 | 09:00 AM – 05:00 PM | NSTI-R, CSA Lab & CHNM Lab

QUICK RECAP

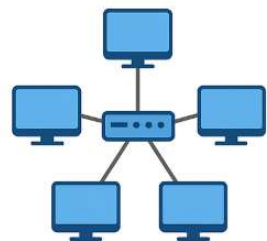
Where We Left Off — Day 2

- **Network topologies:** star, bus, ring, mesh — how devices connect.
- **Structured cabling:** CAT6, crimping, patch panels, IO boxes, racks.
- **Switches & HUBs:** how traffic moves inside a LAN.

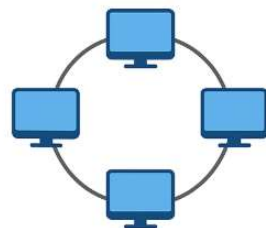
Today's twist: every topology diagram assumes every connected device is trustworthy. What if it isn't?

QUICK RECAP

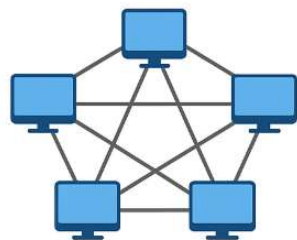
Star Topology



Ring Topology



Mesh Topology



Bus Topology



QUICK RECAP

How to Crimp an Ethernet Cable (RJ45)



Cat6 Ethernet cable



RJ45 connectors



Crimping tool with integrated cutter

- 

1 Strip approx ~1 inch of ethernet jacket
- 

2 Untwist and straighten the 8 internal colored wires
- 

3 Arrange the wires in the correct T568B order
White orange green blue white green brown
- 

4 Trim evenly using the cutter
- 

5 Insert the wires into RJ45 connector
- 

6 Crimping connector by using secured, using the crimping tool
- 

7 Test the cable using a LAN tester

Straight vs Crossover Cable

T568B

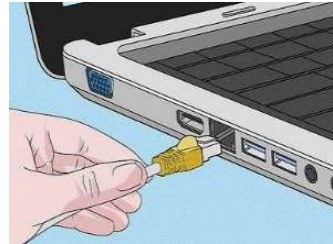
Straight

T568B

T568A

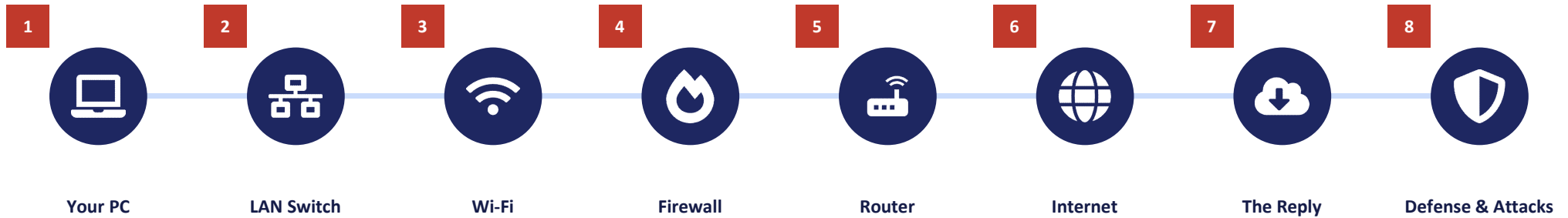
Crossover

T568B



TODAY'S WALKTHROUGH

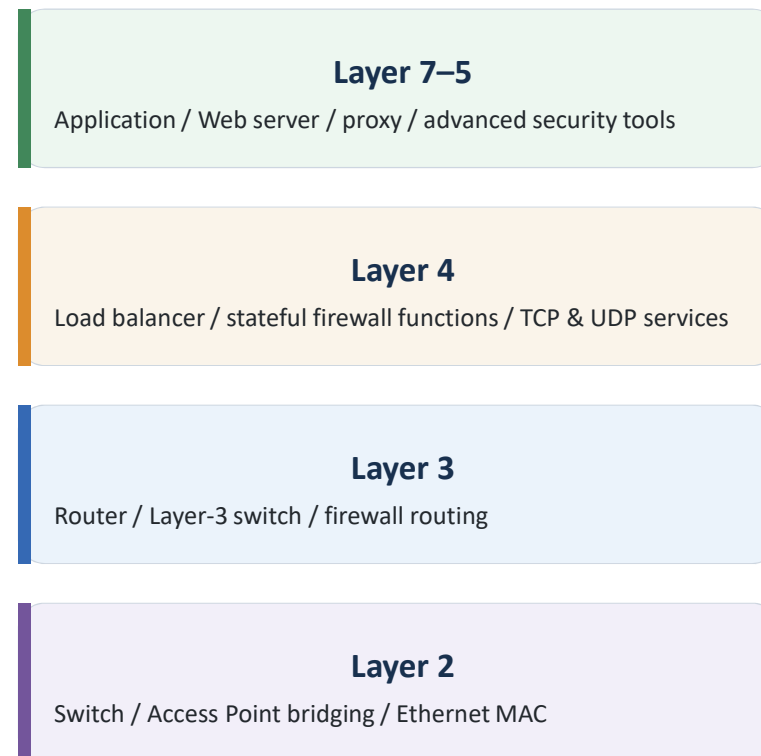
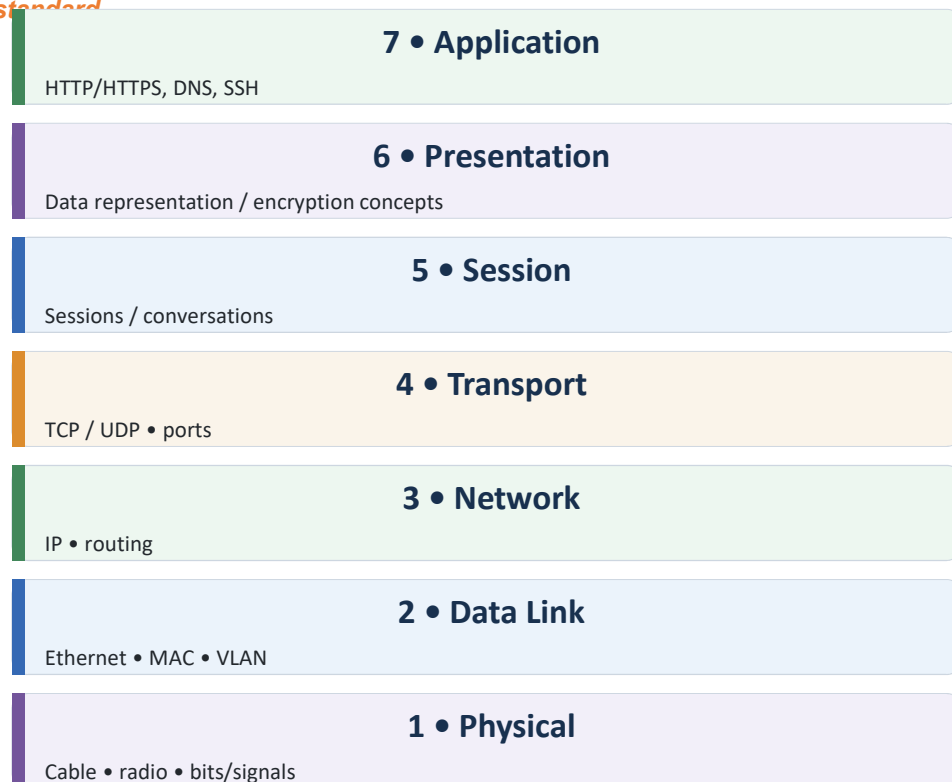
The Journey Map — 8 Steps



We'll trace one journey — your click, all the way to a server and back — then use the same map to see how a live website (built with Python + MySQL, like your Day 4–5 work) defends itself from intruders.

OSI Model: one picture to organize everything

The OSI Model (Open Systems Interconnection model) is a conceptual framework that divides network communication into seven abstract layers. Created by the International Organization for Standardization (ISO), it helps different computer systems and hardware talk to each other by setting a universal standard.



Layer 1 = physical transmission. Layer 2 = local delivery. Layer 3 = routing between networks. Layer 4 = service/application endpoint using ports.

Which device works where? A beginner map

Hub / repeater

Layer 1 • Physical

Regenerates/forwards signals; no MAC intelligence

Switch

Layer 2 • Data Link

Uses MAC address table; forwards Ethernet frames

Access Point

Layer 2 (typically)

Bridges wireless clients into the LAN; Wi-Fi security at the wireless edge

Router

Layer 3 • Network

Routes packets between IP networks

Firewall

Layers 3–7*

Enforces security policy; advanced firewalls inspect higher-layer traffic

IDS / IPS

Often Layers 3–7*

IDS detects/alerts; IPS detects and can block certain malicious traffic

Web Application Firewall

Layer 7

Protects web applications from HTTP/HTTPS application attacks

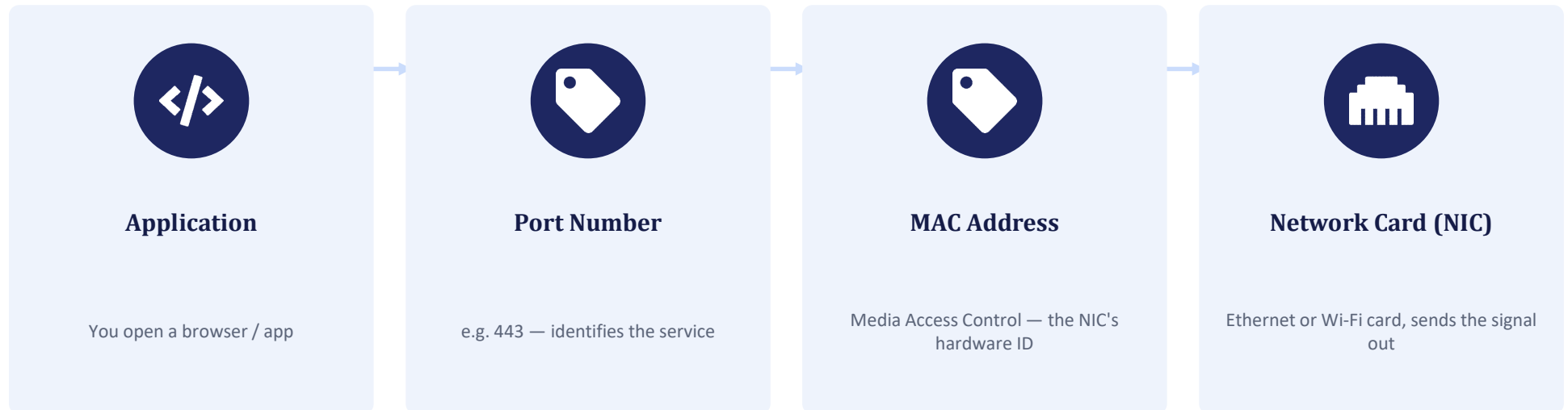
*Exact OSI mapping varies by product. The purpose here is to give students a practical mental model.

REFERENCE CARD

The OSI Model (Open Systems Interconnection) — Layer by Layer

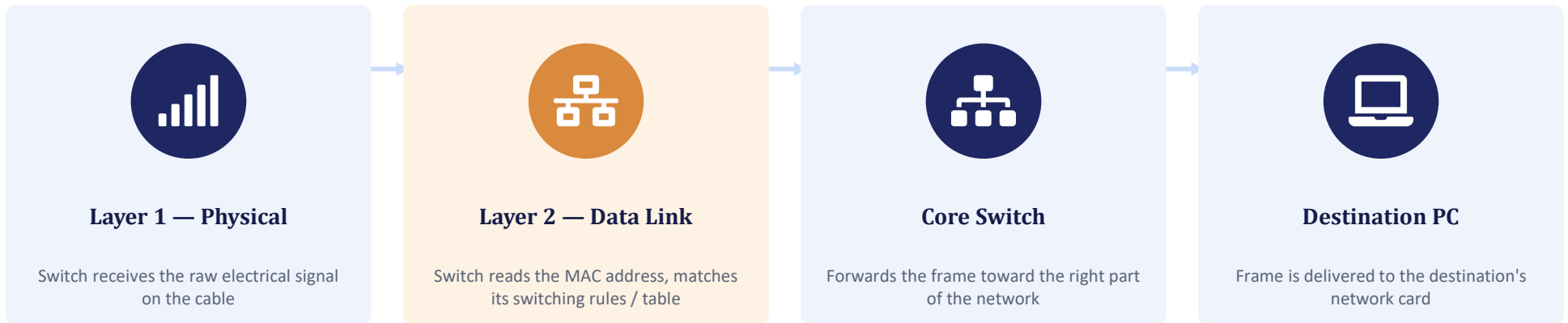
7	 Application	Web browser, email client
6	 Presentation	Encryption/SSL, data formatting
5	 Session	Keeps track of a conversation
4	 Transport	Firewall (stateful), load balancer
3	 Network	Router, Layer-3 switch
2	 Data Link	Switch, network card (NIC)
1	 Physical	Cables, hubs, Wi-Fi radio signal

Inside Your Computer — App → Port → MAC → Network Card



Every layer adds its own “address label” to your data before it leaves the computer — like an envelope inside an envelope.

The Frame Lands on the LAN Switch



Callback: this is Port Security, VLANs (Virtual LANs) and DHCP Snooping in action — the Layer 2 defenses from our last session.

Agenda — 4 Building Blocks of Network Security



1. Secure Admin Access

Locking the door to the device



2. LAN & Layer 2 Security

+ Endpoint security



3. Wi-Fi Security

Encryption & hardening

Anchored throughout to what you already built: OS installs (Day 1), cabling & patch panels (Day 2), and database access control (Day 5).

01



SESSION 1

Secure Administrative Access

Locking the door before locking the house

1. Security starts with three questions

Use these throughout the session

Confidentiality

Who should be allowed to see the information?
Example: student database / MySQL records.

Integrity

How do we know information was not changed incorrectly?
Example: marks or application configuration.

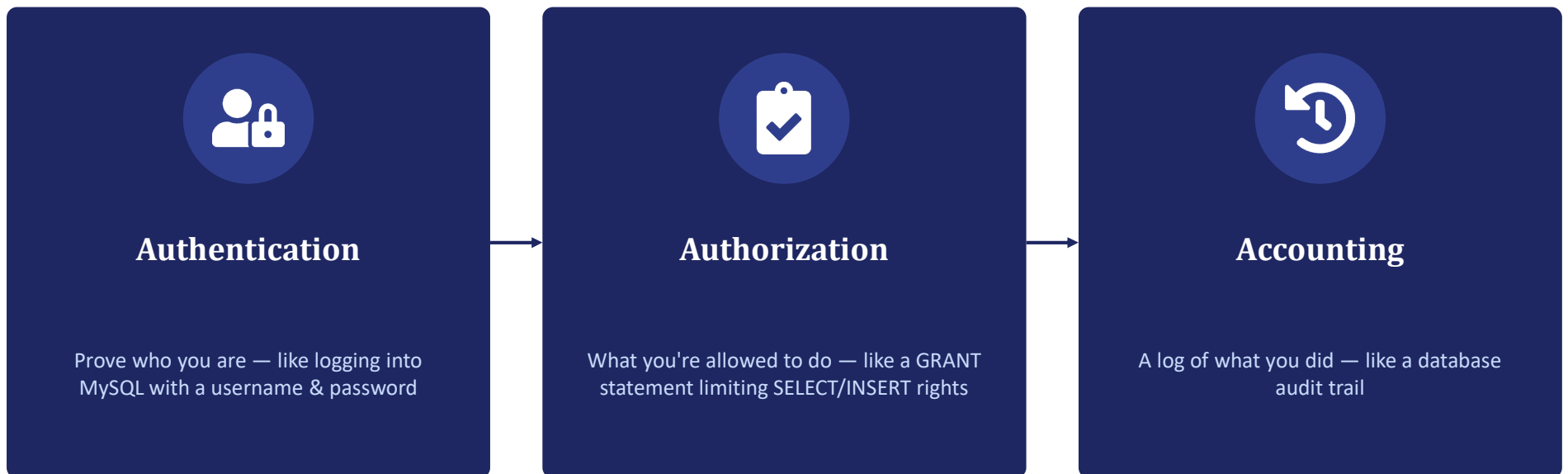
Availability

Can authorized users access the service when needed?
Example: college portal / lab server.

Ask the class: Which one is affected if a website is taken offline?

Answer: availability. If marks are changed: integrity. If passwords are exposed: confidentiality.

AAA — A Callback to Your MySQL Session (Day 5)



“On Day 5 you used GRANT statements to control exactly who could SELECT or INSERT on a table. AAA does the same job for network devices.”

What Is "Administrative Access"?

The different ways an admin logs in to configure a network device — the switch or router you cabled on Day 2.



Console Cable

Direct physical connection — the cable you used in the lab



Telnet / SSH

Remote command-line access over the network



Web GUI

Browser-based device management page

Whoever controls admin access controls the whole network — it's the master key to the building.

Telnet vs SSH — Same Job, Very Different Risk



Telnet

Unencrypted

- Sends everything in plain text — including your password
- Anyone monitoring the network can read the session
- No verification that you're talking to the real device



SSH

Encrypted

- Encrypts the whole session, password included
- Verifies the identity of the device you're connecting to
- The modern standard — always prefer SSH over Telnet

Securing the Door — Best-Practice Checklist



Change default credentials

No device should ever run on factory username/password



Strong passwords + encrypted "enable secret"

Not the plaintext "enable password"



Disable unused services

Turn off Telnet/HTTP if SSH/HTTPS is available



Configure a login banner

Legal notice shown before login



Set an idle session timeout

Auto-logout of forgotten sessions



Restrict access with an ACL

Only specific trusted IPs may manage the device

+ For larger networks: centralize logins with AAA — Authentication, Authorization, Accounting.

Authentication: don't share the Administrator password

Individual accounts

Each administrator has a unique account.
Easier to audit who did what.

MFA where possible

Password + another factor makes stolen
passwords less useful.

Least privilege

Give only the permissions needed for the
task.

Example: a junior lab assistant may restart a service but should not automatically receive full network/device administrator rights.

Use logs to answer: Who logged in? From where? What did they change?



QUICK ACTIVITY

Spot the Risk

```
line vty 0 4
  transport input telnet
  password cisco123
  login
!
enable password admin
!
no service password-encryption
```

- Which lines here are insecure?
- How many issues can your group find in 2 minutes?
- How would you fix each one?

02

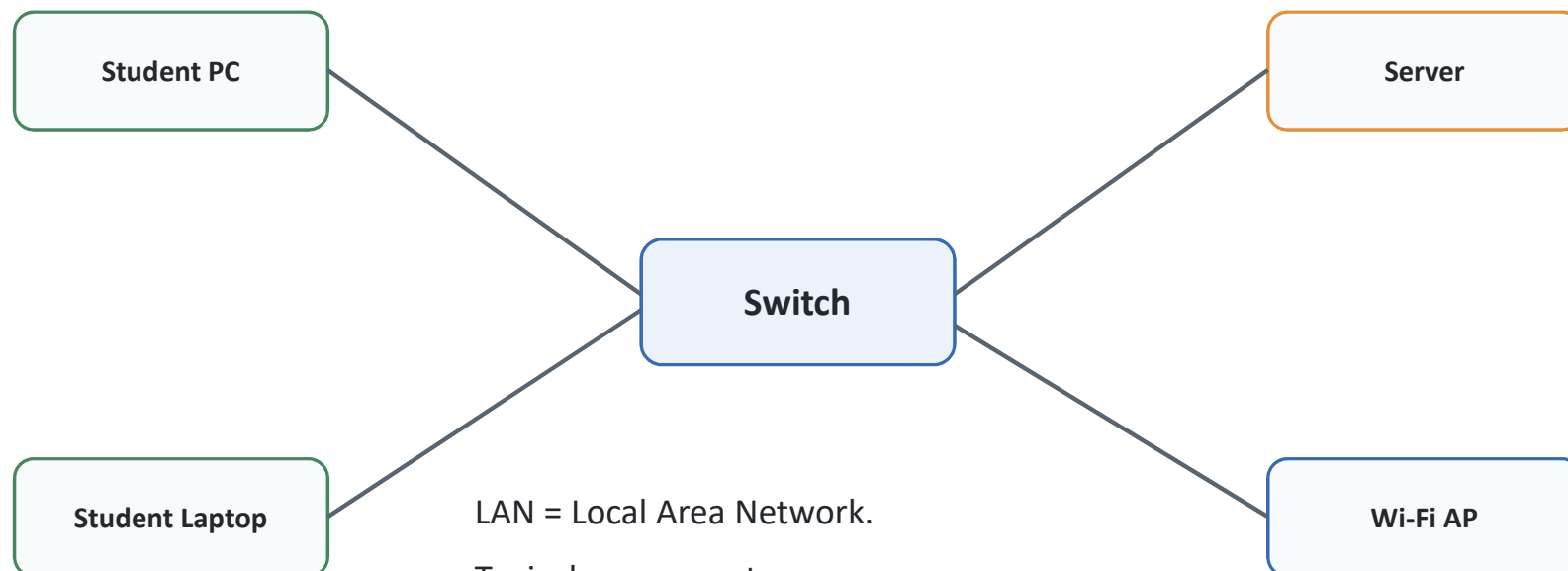


SESSION 2 ·

LAN, Layer 2 & Endpoint Security

Segment, limit, and patch — trust nothing by default

2. What does a LAN look like?



LAN = Local Area Network.

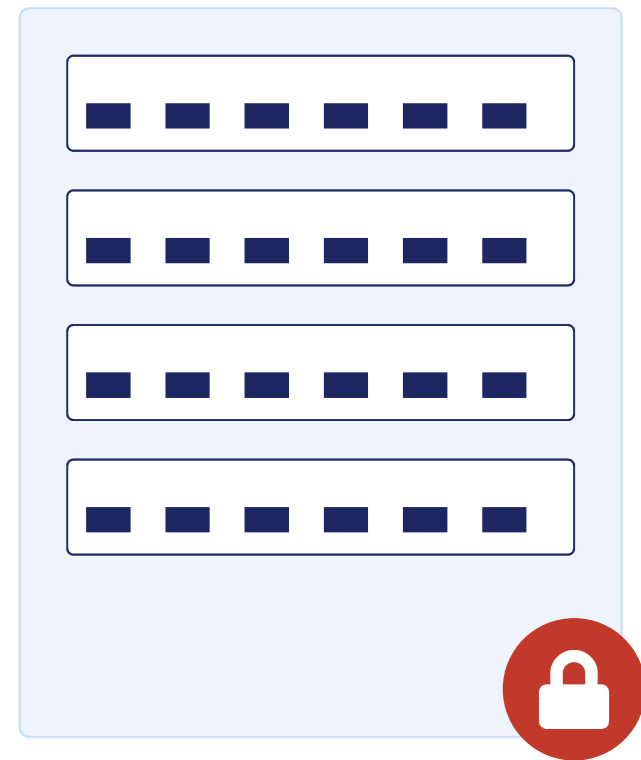
Typical components:
PCs/laptops, switch,
router/firewall, server and Wi-Fi
access point.

Security must protect both the
devices and the traffic moving
between them.

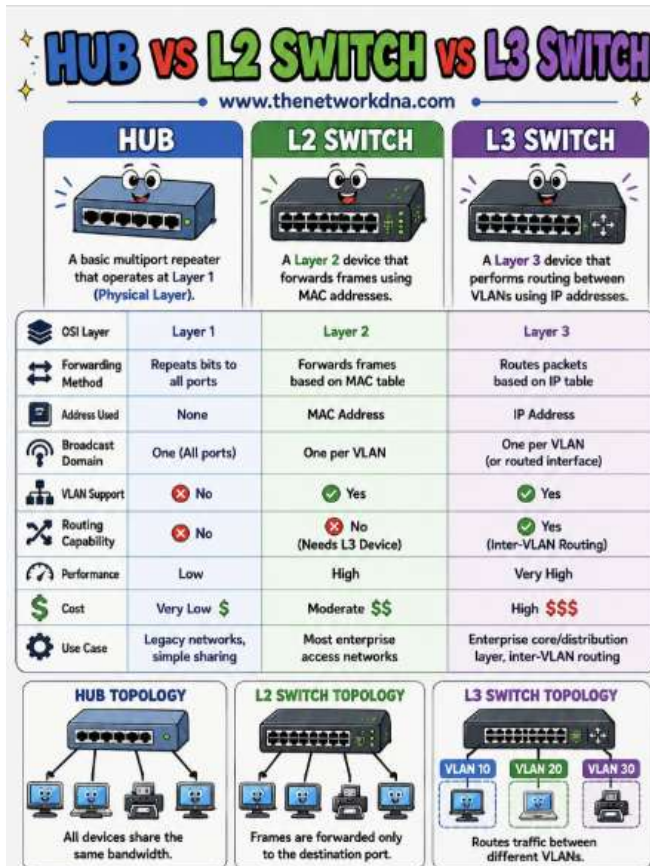
Physical Security Is Network Security

- **Insider risk:** an attacker doesn't need to be outside — a plugged-in rogue device on your own LAN is a real threat.
- **An unlocked server room** is as much a security gap as a software one.
- **Unused switch ports** should be disabled or physically capped.

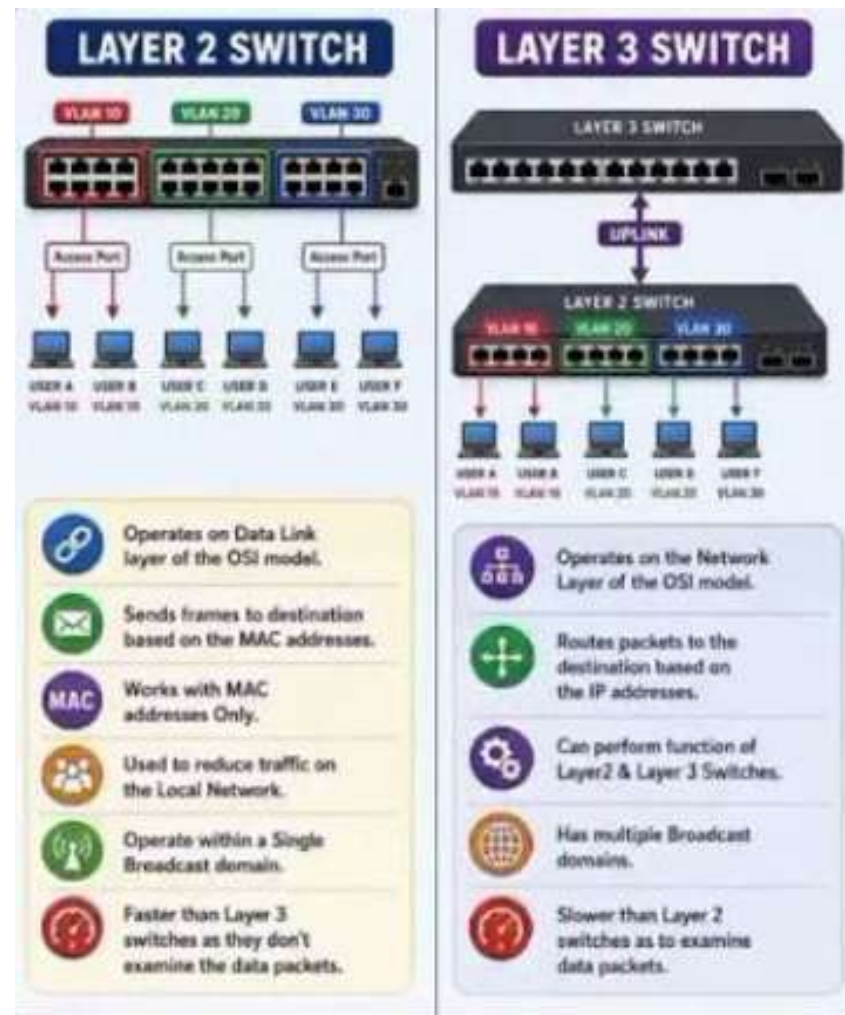
Day 2 callback: the patch panel and rack you cabled — in a real deployment, that rack sits in a locked room, every cable labelled, every unused port capped.



QUICK RECAP



Welcome & Recap



PMSETU IT — Day 8

A Quick Refresher: What Is "Layer 2"?

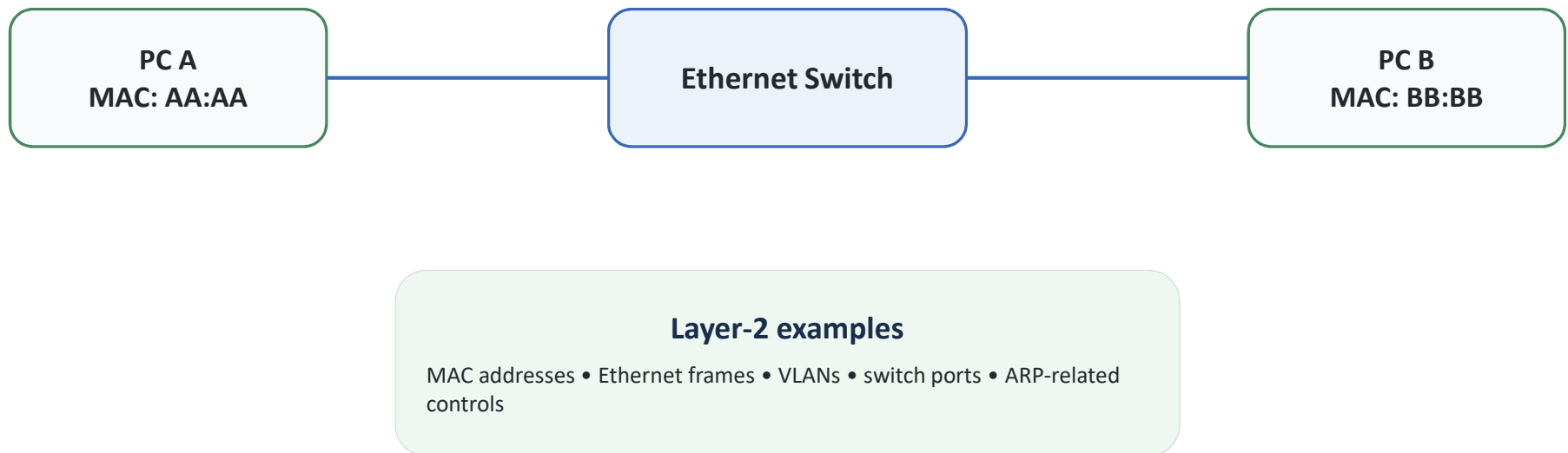


Layer 2 is the switch / MAC-address world — one step below the IP addressing world. It's where your switches, patch panels and cabling from Day 2 do their work.

Because a LAN trusts devices "inside the fence" by default, Layer 2 is a favourite target — and today we look at 4 common attacks, and the defense built for each.

Layer 2: where the switch operates

Simple mental model



Layer 2 is close to the local network connection.

Today we focus on practical controls rather than protocol theory.

Threats and Their Matching Defense



MAC Flooding

Floods the switch's MAC table until it starts broadcasting everything, like a hub



Port Security

Limits how many/which MAC addresses a port accepts



ARP(Address Resolution Protocol) Spoofing

Tricks two devices into sending traffic to the attacker instead of each other



Dynamic ARP Inspection

Validates ARP messages against trusted data



Rogue DHCP(Dynamic Host Configuration Protocol) Server

A rogue device hands out fake IP addresses / a fake gateway



DHCP Snooping

Only trusted ports may act as a DHCP server



VLAN (Virtual LAN) Hopping

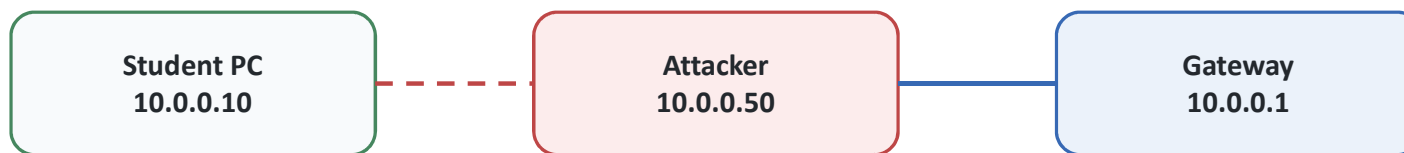
An attacker jumps into a VLAN they shouldn't have access to



VLAN Segmentation

Separate broadcast domains; blocks rogue switches from joining

ARP spoofing – a simple picture



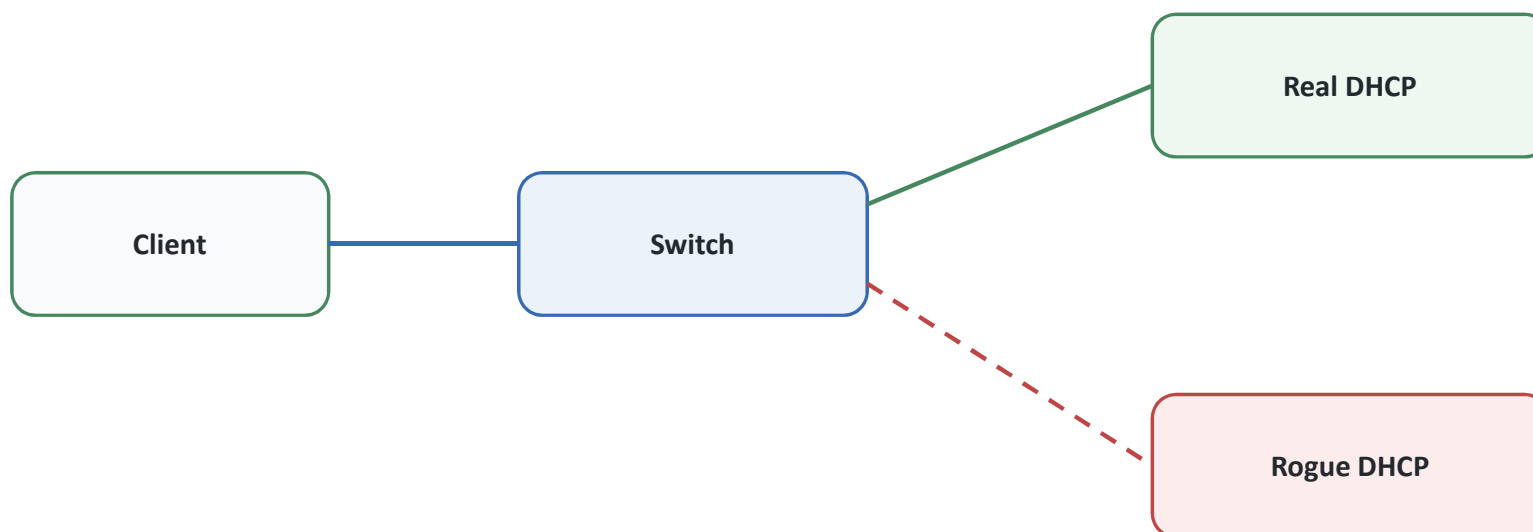
What the attacker tries

Convince a local device that the attacker's MAC address belongs to the gateway (or another host). Traffic may then pass through the attacker.

Defensive idea

Dynamic ARP Inspection (DAI) + DHCP Snooping can help validate ARP information.

DHCP Snooping – stop the fake DHCP server



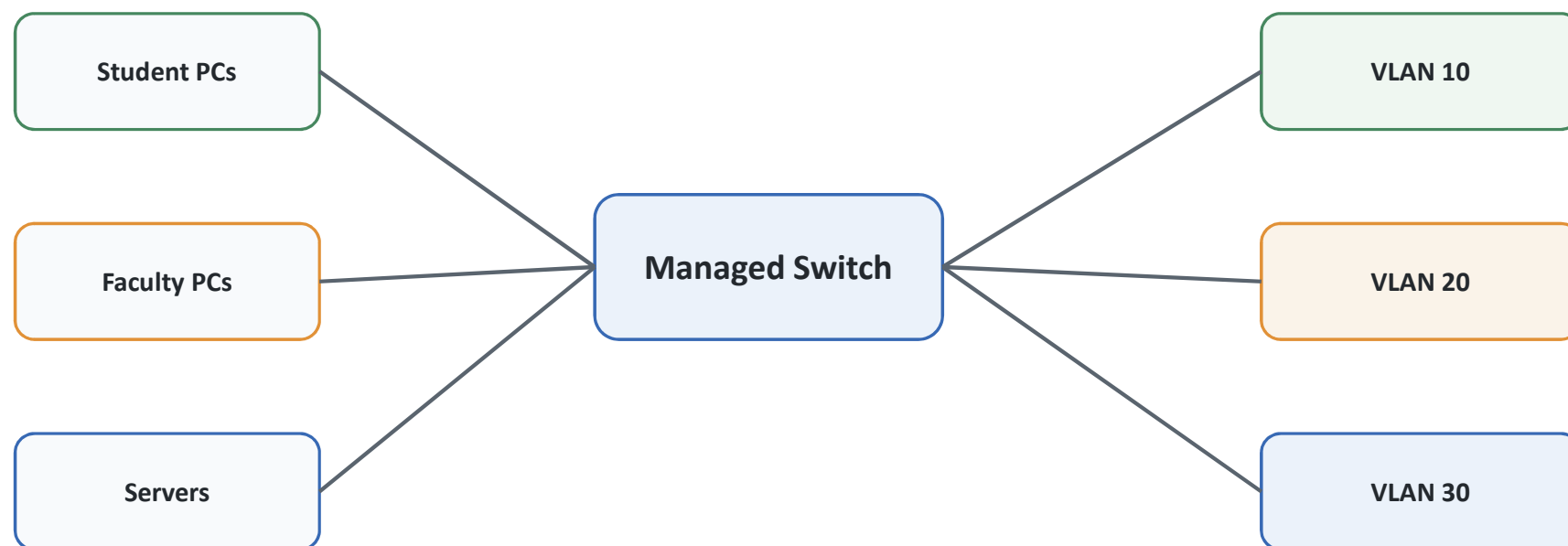
Idea

Configure trusted switch ports toward the legitimate DHCP server/uplink. Unexpected DHCP server messages from untrusted ports can be blocked.

Question: What could a rogue DHCP server give a client?

Possible answers: wrong gateway, wrong DNS server, or other malicious network settings.

VLANs: security through separation

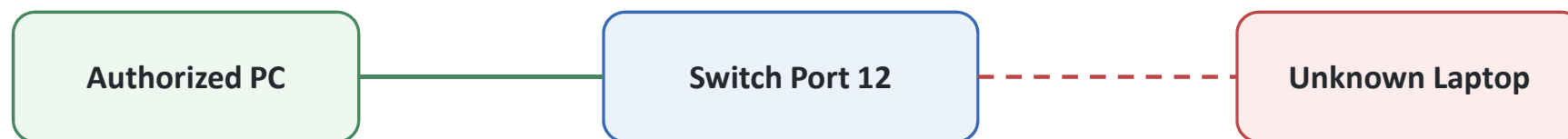


A VLAN is a logical network segment.

Example: students should not be placed in the same broadcast domain as critical servers.

Important: VLAN separation alone does not automatically mean traffic is fully isolated; routing/firewall policy may still be required.

Port security – the simplest story



Concept

The switch can associate a port with an allowed MAC address (or a small number of addresses). If an unexpected device appears, the port can be restricted/shut down depending on the configured violation action.

Example: a lab PC is assigned to Port 12. A student plugs in another laptop using the same port.

Discuss: Why is this useful even though it is not a complete security solution?

What If the Computer Itself Is Infected?



Antivirus / Anti-malware

Scans files and processes for known malware signatures



EDR (Endpoint Detection & Response)

Watches for suspicious behavior, not just known signatures



NAC (Network Access Control)

Can automatically quarantine the infected device off the network



IDS / IPS (Intrusion Detection / Prevention System)

Monitors network traffic patterns for signs of infection spreading

Securing the Endpoint — Your Day 1 Laptop

An "endpoint" is any device that connects to the network — the very laptop you installed Windows/Linux on during Day 1.



Keep the OS patched & updated

Day 1 callback: that Windows/Linux image needs regular security patches



Run antivirus / anti-malware

Detects and blocks known threats



Authenticate devices before they join

802.1X / Network Access Control (NAC), conceptually



Least-privilege user accounts

Day 5 callback: same idea as limited MySQL user privileges



GROUP DISCUSSION

“If you were the IT admin, what would you check on a new laptop before letting it join the office Wi-Fi?”

Think back to the endpoint checklist — patching, antivirus, authentication, least privilege.

03



SESSION 3 ·

Wi-Fi Security Considerations

No cable required — and that changes everything

Why Wi-Fi Is a Different Problem



Wired LAN

An attack needs physical access to a cable or patch panel — the ones you wired on Day 2.



Wi-Fi

The signal travels through walls, out the window, into a parking lot — no physical connection needed.

Common Wi-Fi Threats



Rogue Access Point(AP)

An unauthorized AP plugged into the network, bypassing all your defenses



"Evil Twin"

A fake AP mimicking a real one's name to trick users into connecting



Eavesdropping

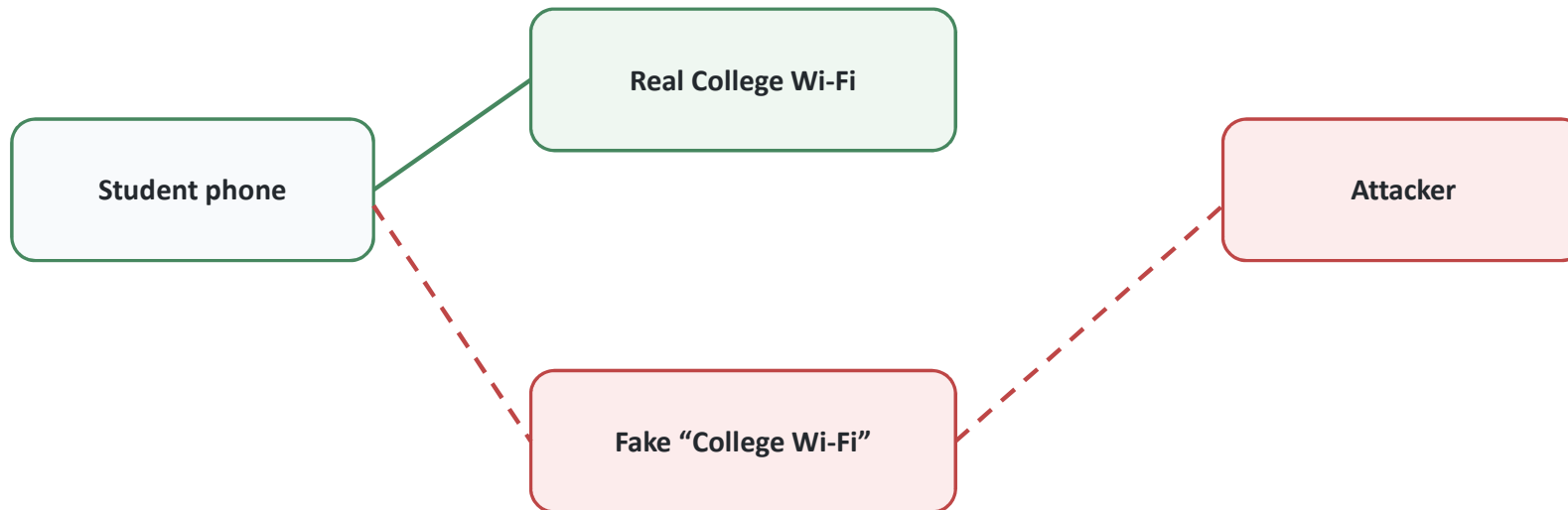
Sniffing traffic on open, unencrypted Wi-Fi networks



Weak / Default Passwords

The single most common real-world Wi-Fi breach

Rogue AP / Evil Twin – why Wi-Fi users must be careful



Lesson

Users should verify the network, use secure authentication, and avoid entering sensitive credentials into suspicious captive portals or unknown networks.

Wi-Fi security: the basics

Use modern encryption

Prefer WPA3 where supported; otherwise WPA2 with strong configuration is common.

Strong authentication

Use strong unique credentials; for organizations, enterprise authentication can provide per-user access.

Separate guest network

Guest devices should not automatically reach internal servers.

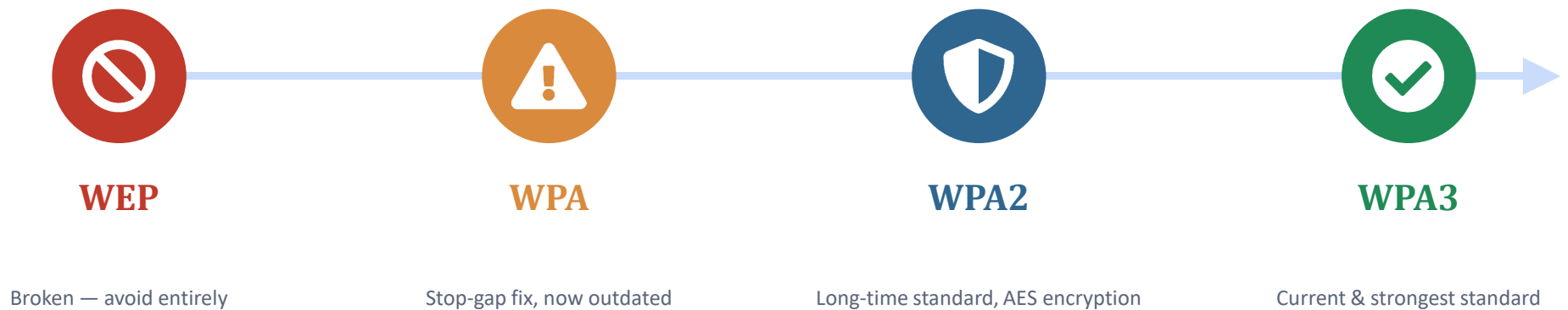
Avoid treating the Wi-Fi password as the only security control.

Change default admin credentials on APs/routers.

Keep firmware updated.

Disable features you do not need.

The Wi-Fi Encryption Timeline



Always deploy WPA2 or WPA3 — never WEP, and avoid WPA-only where WPA2/3 is available.

WEP (Wired Equivalent Privacy): Introduced in 1997 as part of the original 802.11 wireless standard.

WPA (Wi-Fi Protected Access): Released in 2003 by the Wi-Fi Alliance as a quick fix to replace WEP. It uses dynamic keys that change for each packet

Practical Wi-Fi Hardening Checklist

- **Use WPA2/WPA3** with a strong, unique passphrase
- **Change the default** router admin password and SSID (callback to Session 1!)
- **Disable WPS** — a known weak point in many routers
- **Separate Guest network / VLAN** for visitor devices (callback to Session 2's VLANs)
- **Keep firmware updated** (callback to endpoint patching)



Myth-busting

Hiding the SSID and MAC filtering are NOT real security — both are trivially bypassed with free tools. They add friction, not protection.

Network Access Control – who gets onto the network?

Open access

Any device can connect. Easy, but weak control.

Controlled access

Use VLANs, port security or 802.1X to decide who/what connects.

Guest access

Visitors use a separate network with restricted access to internal resources.

Example: college Wi-Fi should not automatically provide the same access as the server VLAN.

Ask: Where would you place guest devices? Student devices? Servers?

Final recap: the security mindset

1 • Control access

Who is allowed to connect and who is allowed to administer?

2 • Separate

Do not put every user and every server into one flat network.

3 • Secure management

Use encrypted protocols, individual accounts, MFA where possible and least privilege.

4 • Protect Layer 2

Port security + DHCP Snooping + DAI + unused-port controls.

5 • Protect Wi-Fi

Modern security, strong authentication, segmentation and monitoring.

One sentence for students: “Security is not one product — it is a set of controls that reduce the ways an attacker can get in, move around, or misuse systems.”

Recap Quiz — Quick Fire Round



Which protocol encrypts admin sessions — Telnet or SSH?

Q1



What does Port Security limit on a switch port?

Q2



Name one Wi-Fi encryption standard you should avoid.

Q3



What network segment keeps guest Wi-Fi users away from internal systems?

Q4

Mini lab: design a secure college LAN

Draw four zones: Student, Faculty/Admin, Servers and Guest.

Place at least one switch, firewall/router and Wi-Fi AP.

Choose where VLANs are used.

Mark which switch ports should be trusted for DHCP.

Decide how administrators reach the devices.

Write 3 firewall rules.

Add 3 Wi-Fi security controls.

Bonus: identify one possible attack and the control that reduces it.

Deliverable

One-page network diagram + 6–10 security decisions.

Team time

15 minutes

Presentation

2 minutes per team

Case study: the insecure lab

A lab has 30 PCs on one flat LAN.

The switch has 8 unused ports, all active.

Everyone uses the same Wi-Fi password.

The switch is administered from any PC on the LAN.

A student plugs in a personal router and suddenly some users get the wrong gateway.

The MySQL server is reachable from every PC.

Find the problems

At least 6 weaknesses are present.

Fix them

Use concepts from today: VLANs, port security, DHCP snooping, admin restriction, firewalling and Wi-Fi controls.

TODAY'S GOLDEN RULES

Five Ideas Worth Remembering



Secure the door before you secure the house

Lock down administrative access first



Segment and limit

VLANs, port security, DHCP snooping — only allow what you expect



Patch the endpoint, not just the network

The laptop matters as much as the switch



WPA2/WPA3 + a strong passphrase

Beats every "hidden SSID" trick



Physical security is network security

Locked racks and capped ports still matter

Callbacks we used today

- Day 1 — OS installs → endpoint patching
- Day 2 — cabling & patch panels → physical & LAN security
- Day 5 — MySQL GRANTS → AAA & least privilege

Exit question

Ask every student to answer before leaving

“If you were given a college computer lab tomorrow, what are the FIRST 5 security controls you would check?”

Expected themes: admin access, VLAN/segmentation, unused ports, DHCP/Wi-Fi controls, firewall/access rules, updates/logging.

End with a 2-minute student reflection: one concept they understood well + one concept they want to practice.

Network Engineer

"I make sure everything connects."



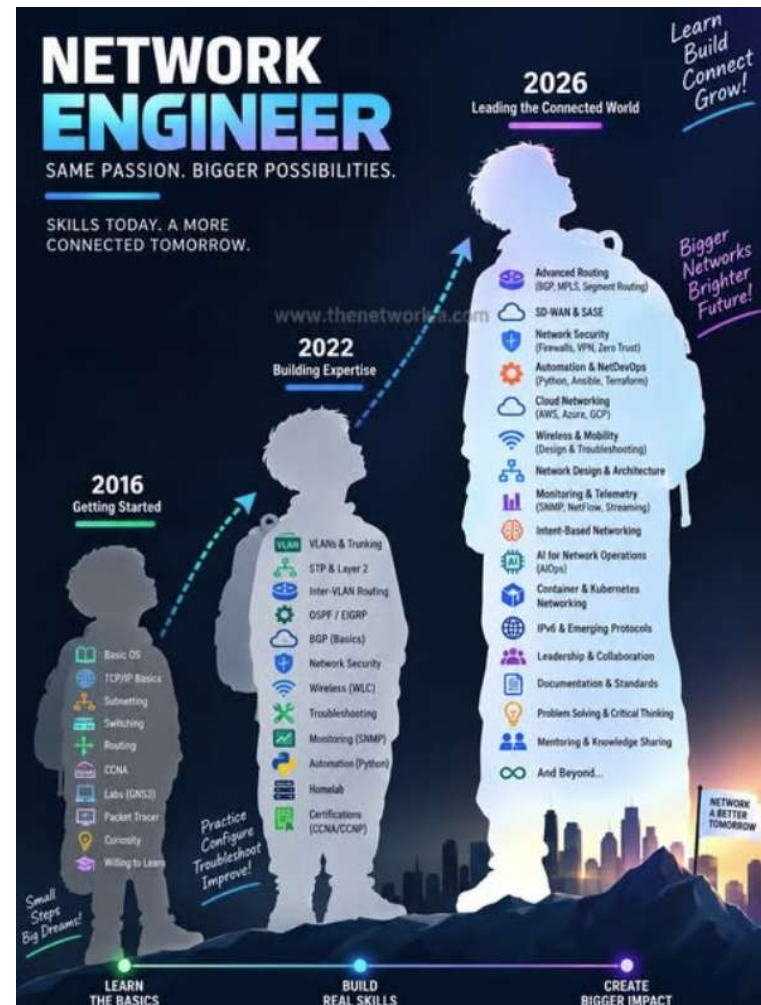
"If it's not working... I'll find the path!"

Cybersecurity Professional

"I make sure the bad guys don't connect."



"Same network... Different mission!"





Thank You

Questions & Open Discussion